

Genetic Disease Inheritance Probability Analysis Using Random Forest Classification

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Abstract—Genetic diseases such as *Thalassemia, Hemophilia, Breast Cancer, Sickle Cell Anemia, and Cystic Fibrosis* pose significant health risks that can be inherited from offspring. Early identification of genetic disease risk is essential for preventive medical measures. This paper presents a web-based application that predicts the probability of inheritance of a genetic disease using a Random Forest classification model trained with a dataset of 1,000 clinical records with 12 features. The dataset is divided into 70% training and 30% testing. The model achieved an accuracy of 94%, with precision, memory and F1 scores all averaging 94% across five disease classes. This application calculates the risk of inheritance of a child based on Mendel's Law on accompanying inheritance patterns, including Autosomal Recessive, X-linked Recessive, and Autosomal Dominant/Multifactorial. The results show that the proposed system can reliably identify disease patterns from individual clinical data and provide meaningful inheritance risk estimates.

Keywords—genetic disease, inheritance, random forest, Mendel's Law, classification

I. INTRODUCTION

Genetic diseases are medical conditions caused by abnormalities in a person's genome. Unlike infectious diseases, genetic conditions can be passed from parents to their children through hereditary mechanisms. Diseases such as Thalassemia, Hemophilia, Breast (hereditary) Cancer, Sickle Cell Anemia, and Cystic Fibrosis represent some of the most common congenital conditions globally, affecting millions of families.

A major challenge in the management of genetic diseases is the lack of accessible tools that allow individuals to understand whether they carry genetic risk factors and what likelihood their children may inherit the condition. Traditional genetic counseling is expensive, and time-consuming.

Recent advances in machine learning have enabled the development of predictive models capable of identifying disease patterns from clinical data. One of these is Random Forest, an ensemble learning method, which has shown strong performance in medical classification tasks due to its robustness to noise and its ability to handle high dimensional table data.

II. LITERATURE REVIEW

2.1 Machine Learning and Random Forest in Genomic Classification

Several studies have explored the application of machine learning, most notably Random Forest, in medical diagnostics and predictive of inherited conditions. In genomics, handling high-dimensional data is a well-known computational challenge. Previous research by Ma. W et al. (2022) showed that the Random Forest algorithm is very robust in this domain, successfully constructing a predictive genetic risk model for systemic lupus erythematosus from whole genome association data. By utilizing an ensemble of decision trees, this architecture effectively captures

non-linear genetic effects, maintaining predictive accuracy even when informative genetic features are scarce. In addition, a study by Mukherjee. S et al. (2021) prove that machine learning frameworks have been shown to be highly capable of identifying complex multigenetic interactions, such as finding digenic disease genes in undiagnosed disease network .

2.2 Flexibility of Random Forest in Epidemiological and Clinical Setting

The Random Forest algorithm has been extensively validated in a wide range of clinical classification tasks. A study by Özen. F (2024) confirms that this algorithm has been successfully applied to predict epidemiological trends such as COVID-19 cases. Also, study by Tran. Hoc et al (2023) and Tang. Y et al (2024) proves that in pathogen classification, Random Forest can be used to categorize rotavirus genotypes with test accuracy exceeding 97% and trace the source of Salmonella Typhimurium isolates, confirming its adaptability at handling complex biological data.

2.3 Predictive Modeling For Specific Genetic Diseases

For specific hereditary diseases, computational models show remarkable precision. Research by Khandakji. N. M and Mifsud. B (2022) evaluated the pathogenicity of BRCA2 variants, directly informing our Breast Cancer prediction module using machine learning. In blood clotting disorders, Rawal. A et al. (2024) predicted the severity of Hemophilia A by analyzing genetic mutations. For blood disorders, Abbas. M et al. (2025) and Gómez. G. J et al. (2025) proved that this architecture can predict the status of Thalassemia carrier with 99.30% accuracy and classify the anemia system. Moreover, a study by Qin. Y et al. (2023) validated that machine learning-based prognostic scores are highly effective in predicting clinical outcomes for Cystic Fibrosis. These works collectively confirm the feasibility of using machine learning to classify the five diseases in this study

2.4 Integration of Predictive Modeling with Rules-Based Clinical Interpretation

Although Random Forest provides high accuracy, clinical applications require interpretability beyond numerical output. To address the nature of the black box inherent in machine learning models, Abbas. M et al. (2025) demonstrate the value of combining Explainable AI (XAI) techniques, specifically SHAP and LIME to provide transparent clinical justification for model predictions. Inspired by this research, our system integrates a structured rule-based explanation layer directly into the prediction flow. Once the Random Forest classifier identifies a genetic condition, the system programmatically points the results to a predefined clinical explanation and calculates the feature importance score of the model. This hybrid approach ensures that each prediction is accompanied by a medical explanation and a list of the most influential parameters, making the system clinically interpretable without the need for an external XAI library.

2.5 Bridging Algorithmic Classification with Mendelian Inheritance

Although significant progress has been made in predicting disease from clinical data, the translation of individual-level predictions into family-level inherited risk

remains an uncharted area. Mendel's laws of inheritance such as Autosomal Recessive, Autosomal Dominant, and X-linked Recessive remain the theoretical cornerstones of genetic counseling and define how pathogenic variants multiply across generations. Previous study by Mukherjee. S et al. (2021) and Khandakji. N. M and Mifsud. B (2022) focuses on the identification of disease-causing variants at the individual level, none of which directly automate the calculation of the probability of inheritance for offspring. This represents a clear research gap. Our system addresses this by combining the output of the Random Forest classification with a Mendel-based probability engine. Once the patient's disease status is confirmed, the system calculates probabilistic inheritance outcomes for the prospective child based on the pattern set for each disease.

III. METHODOLOGY

The dataset used in this study was obtained from Kaggle and contains 1,000 individual clinical records. Each record consists of 12 clinical features and one label column indicating the associated genetic disease. Each disease is distributed with 200 records, ensuring a balanced classification task .

Table 3.1 Dataset Summary

Attributes	Value
Total Records	1,000
Number of Features	12
Number of Classes	200
Records per Class	200
Training Split	70% (700 records)
Testing Split	30% (300 records)

The clinical features that are used were age, gender, family history, Hemoglobin, Fetal Hemoglobin, RDW-CV, Serum Ferritin, BRCA1 Expression, p53 Mutation, Sweat Chloride, Sickle RBC, and IL-6 Level. These features were selected based on their clinical relevance to each of the five target diseases.

Table 3.2 Clinical Features Range

Features	Range
Gender	Female-Male
Family History	None-Yes
P53 Mutation	None-Present
Age	10-69
Hemoglobin (g/dL)	5-18
Fetal Hemoglobin	0-30
RDW-CV	10-30

Serum Ferritin (ng/mL)	5-500
BRCA1 Expression	0-10
Sweat Chloride (mmol/L)	10-120
Sickled RBC	0-10
IL-6 Level (pg/mL)	0-35

Two preprocessing steps are used for the dataset. First, the Feature Matrix (X) and the Label Vector (y) are separated, isolating then making the Disease column as the classification target. Then, Standard Scaling (Z-score normalization) is applied to all 12 features to ensure equal contribution across all features with different numerical ranges. Although Random Forest is relatively insensitive to feature scale, standardization is still applied to maintain preprocessing consistency and facilitate future use of other algorithms.

A Random Forest Classifier was chosen as the primary model due to its established performance on tabular clinical data, resistance to overfitting, and ability to produce feature importance scores. The model was configured with 200 decision trees ($n_{\text{estimators}} = 200$) and a maximum depth of 10. The scikit-learn library was used for the implementation. Then, to preserve class balance the dataset was split into 70% training and 30% testing using stratified sampling. The trained model along with the scaler was serialized using joblib for deployment.

After predicting the patient's disease status, the system calculates child inheritance probability based on Mendel's Laws. Three patterns of inheritance are applied:

1. Autosomal Recessive for Thalassemia, Sickle Cell Anemia, and Cystic Fibrosis. Both parents must contribute a defective allele for a child to be affected.
2. Autosomal Dominant for Breast Cancer. One copy of the BRCA1 variant significantly elevates the risk.
3. X-linked Recessive for Hemophilia. The gene is carried on the X chromosome, making males more susceptible.

The app is divided into two components. The backend is Python Flask REST API that loads trained models and exposes predictive endpoints. The frontend is React Javascript that provides a clinical data input interface and visualizes predictive outcomes. The system is also deployed , using Railway for the backend and Vercel for the frontend.

IV. RESULT AND DISCUSSION

The Random Forest model achieved an overall accuracy of 94% on a test set of 300 patients records.

Table 4.1 Classification Results

Disease	Prec.	Rec.	F1-score	Sup.
Thalassemia	0.93	0.95	0.94	59

Hemophilia	0.93	0.93	0.94	59
Breast Cancer	0.97	0.93	0.95	61
Sickle Cell	0.95	0.93	0.94	61
Cystic Fibrosis	0.92	0.95	0.93	60
Average / Total	0.94	0.94	0.94	300

This model shows balanced performance across all five classes of diseases. Breast cancer achieves the highest precision, attributed to distinctive BRCA1 expression and p53 mutation features. Meanwhile, hemophilia shows the lowest precision, potentially due to the overlapping hematologic profile with Thalassemia. Overall, Random Forest is suitable for this multi-class genetic disease classification task with 94% accuracy.

In addition, Feature Importance is also analyzed using the Mean Decrease in Impurity (MDI) mechanism using Random Forest's built-in feature in scikit-learn. Whenever a feature is used to split nodes in the decision tree and successfully reduce impurity, it receives a value contribution. The contributions of the entire tree are then summed and normalized to produce the importance of each feature.

Table 4.2 Feature Importance

Feature	Importance
Sickled RBC Percent	16.3%
BRCA1 Expression	13.5%
Sweat Chloride	13.4%
Fetal Hemoglobin	13.3%
IL6 Level	12.7%
Serum Ferritin	12.4%
Hemoglobin	8.1%
RDW CV	5.6%
p53 Mutation	2.6%
Age	1.1%
Family History	0.2%
Gender	0.2%
Accuracy	94%

The Feature Importance analysis revealed that the Sickled RBC Percent, BRCA1 Expression, Sweat Chloride, Fetal Hemoglobin, and IL6 Level were the five most influential features. These results are in line with established

clinical indicators, where Sickled RBC is a direct marker for Sickle Cell Anemia, BRCA1 is a biomarker for hereditary breast cancer, and Sweat Chloride is the primary diagnostic for Cystic Fibrosis.

Confusion Matrix is also used to evaluate the performance of classification models in more detail. In contrast to accuracy, which only shows the overall percentage of correct predictions, the Confusion Matrix also shows which classes are correctly predicted and which classes are confused with other classes.

Table 4.3 Confusion Matrix

Actual / Pred	Thal	Hemo	BC	SCA	CF
Thal	56	1	0	0	2
Hemo	1	55	1	2	0
BC	1	1	57	0	1
SCA	1	2	0	57	1
CF	1	0	1	1	57

The model showed strong results in all five classes. Thal (Thalassemia) achieved 56 correct predictions out of 59, with 2 being misclassified as CF (Cystic Fibrosis) likely due to the overlapping hematologic profile between the two conditions, as both affect red blood cells. Hemo (Hemophilia) correctly classified 55 out of 59 samples, with 2 misclassifications against SCA (Sickle Cell Anemia), which may reflect a shared low hemoglobin indicator. BC (Breast Cancer) and SCA (Sickle Cell Anemia) each resulted in 57 correct predictions, with BC (Breast Cancer) 2 misclassified directed to CF (Cystic Fibrosis), suggesting that certain BRCA1 expression levels may share a feature pattern with CF (Cystic Fibrosis) markers. CF (Cystic Fibrosis) also achieved 57 correct predictions out of 60, with 3 misclassifications.

There is no single class that showed a dominant misclassification direction, suggesting that the model did not show a systematic bias toward a particular disease. The overall pattern confirms that 94% accuracy is consistent across all five classes.

Finally, some limitations must be noted. The datasets used are synthetic but are realistic, making them suitable for training robust machine learning models for genetic disease prediction, but may not fully reflect real-world clinical complexity. The carrier/affected status threshold is determined by a fixed confidence percentage of the model's output, which is a simplification of actual genetic testing. The inheritance calculation assumes a single gene disorder, which can simplify multifactorial several conditions.

V. CONCLUSION

This paper presented a web-based Genetic Disease Inheritance Analyzer that combines Random Forest classification with Mendelian's probability calculation/ The system was trained on 1,000 clinical records across five genetic diseases and achieved 94% classification accuracy with 70/30 train-test split. The application successfully predicts disease association from individual clinical parameters and estimates child inheritance risk based on established genetic inheritance patterns.

This system demonstrates that machine learning can be integrated with genetic theory to produce accessible clinical decision support tools. The deployed web application (Genetic Disease Analyzer) which can be accessed at <https://geneticanalyzer.vercel.app/> provides real-time analysis with interpretable results, including the most influential factors for the prediction.

VI. AI Acknowledgement

Generative AI (Gemini) was used to assist with language refinement and preliminary summarization during manuscript preparation. All AI-generated outputs were critically reviewed and revised by the author(s). The author(s) assume full responsibility for the accuracy, originality, and content of this manuscript.

VII. OPEN DATA

The data and results of data processing of this research can be accessed through the following link: <https://doi.org/10.5281/zenodo.20679482>

VIII. AUTHOR CONTRIBUTION

Marco Alexius: App idea, created the Genetic Disease Analyzer web based application, research paper, dataset, provides data for presentation, app deployment. **Ignatius Roger Ramopolii:** App idea, app Interface prototype, present the presentation, presentation materials, dataset. **Vincentius Henryanto:** Present the presentation, app idea. **Devano Carlo Putra Anta:** Presentation, presentation material, app idea. **Fiki Ramadhan:** Present the presentation, presentation making, presentation material. **Gede Bhoja Naradhipa:** App idea, presentation, presentation material, present the presentation, dataset. **Clement Jonatan Agustian Tanjaya:** Presentation material, presentation, present the presentation. **Dewa Ayu Defina Audrey Nathania:** Review and Supervise.

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